1–Forms

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Abstract. This study note reviews the algebraic and geometric foundations of 1-forms on Euclidean 3-space ($\mathbf{E}^3$) following the framework of Barrett O'Neill. By establishing the behavior of the differential operator d over functions and tangent vectors, we provide the necessary exterior calculus toolset to interpret classical electrodynamics. We then apply these tools to analyze how the vector and scalar potentials undergo local gauge transformations while preserving the invariance of physical observables.

Contents

1. The Euclidean Space

In the context of differential geometry, we tend to work within the *Euclidean space*. Evidently, it's only useful to define it for future directions.

Definition 1.1. *Euclidean* 3*-space* $\mathbf{E}^3$ is the set of all ordered triples of real numbers. Such a triple $\mathbf{p} = (p_1, p_2, p_3)$ is called a *point of* $\mathbf{E}^3$.

2. 1–Forms

If we have a real–valued function $f \in \mathbf{E}^3$, recall that differentiability is defined as:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz. \tag{1}$$

Definition 2.1. A 1*-form* ϕ *on* $\mathbf{E}^3$ is a real-valued linear function defined on the set of all tangent vectors at each point, satisfying $\phi(a\mathbf{v} + b\mathbf{w}) = a\phi(\mathbf{v}) + b\phi(\mathbf{w})$ for tangent vectors $\mathbf{v}, \mathbf{w}$ at the same point.

The entire equation shows that the function ϕ preserves this structure. This property is called linearity (or a linear transformation). This means you get the same result whether you combine the vectors first and then apply ϕ, or apply ϕ to each vector individually and combine them after.

The algebraic operations on 1-forms are defined pointwise as follows:

- **Addition**: The sum of 1-forms ϕ and ψ is defined by
$$(\phi + \psi)(\mathbf{v}) = \phi(\mathbf{v}) + \psi(\mathbf{v}) \tag{2}$$
for all tangent vectors $\mathbf{v}$.
- **Function Multiplication**: If f is a real-valued function on $\mathbf{E}^3$ and ϕ is a 1-form, the product $f\phi$ is defined by
$$(f\phi)(\mathbf{v_p}) = f(\mathbf{p})\phi(\mathbf{v_p}) \tag{3}$$
for all tangent vectors $\mathbf{v_p}$ at point $\mathbf{p}$.

A 1-form ϕ evaluates on a vector field V pointwise to yield a real-valued function $\phi(V)$, defined at each point $\mathbf{p}$ by:
$$\phi(V)(\mathbf{p}) = \phi(V(\mathbf{p})) \tag{4}$$

This evaluation mapping is bilinear over functions (meaning it is linear in both arguments with respect to scalar functions f and g):

- **Linearity in V**:
$$\phi(fV + gW) = f\phi(V) + g\phi(W) \tag{5}$$
- **Linearity in ϕ**:
$$(f\phi + g\psi)(V) = f\phi(V) + g\psi(V) \tag{6}$$

3. Converting Functions into 1–Forms

Recall the following:

Definition 3.1. Let f be a differentiable real-valued function on $\mathbf{E}^3$, and let $\mathbf{v_p}$ be a tangent vector to $\mathbf{E}^3$. Then the number
$$\mathbf{v_p}[f] = \frac{d}{dt}(f(\mathbf{p} + t\mathbf{v}))\bigg|_{t=0} \tag{7}$$
is called the *directional derivative of f with respect to $\mathbf{v_p}$*.

Theorem 3.2. *Let f and g be functions on $\mathbf{E}^3$, $\mathbf{v_p}$ and $\mathbf{w_p}$ tangent vectors, and a, b numbers. Then:*

1. $(a\mathbf{v_p} + b\mathbf{w_p})[f] = a\mathbf{v_p}[f] + b\mathbf{w_p}[f]$.
2. $\mathbf{v_p}[af + bg] = a\mathbf{v_p}[f] + b\mathbf{v_p}[g]$.
3. $\mathbf{v_p}[fg] = \mathbf{v_p}[f] \cdot g(\mathbf{p}) + f(\mathbf{p}) \cdot \mathbf{v_p}[g]$.

Using the notion of directional derivative, we now define a most important way to convert functions into 1-forms.

Definition 3.3. If f is a differentiable real-valued function on $\mathbf{E}^3$, the *differential df* of f is the 1-form such that
$$df(\mathbf{v_p}) = \mathbf{v_p}[f] \quad \text{for all tangent vectors } \mathbf{v_p}. \tag{8}$$

4. Examples of 1–Forms

Example 4.1. 1-Forms on $\mathbf{E}^3$. (1) The differentials dx_1, dx_2, dx_3 of the natural coordinate functions. Using Lemma 3.2 we find

$$dx_i(\mathbf{v_p}) = \mathbf{v_p}[x_i] = \sum_j v_j \frac{\partial x_i}{\partial x_j}(\mathbf{p}) = \sum_j v_j \delta_{ij} = v_i \tag{9}$$

where δ_{ij} is the Kronecker delta (0 if $i \neq j$, 1 if $i = j$). Thus the value of dx_i on an arbitrary tangent vector $\mathbf{v_p}$ is the ith coordinate v_i of its vector part—and does not depend at all on the point of application $\mathbf{p}$.

(2) The 1-form $\psi = f_1\, dx_1 + f_2\, dx_2 + f_3\, dx_3$. Since dx_i is a 1-form, our definitions show that ψ is also a 1-form for any functions f_1, f_2, f_3. The value of ψ on an arbitrary tangent vector $\mathbf{v_p}$ is

$$\psi(\mathbf{v_p}) = \left(\sum_i f_i\, dx_i\right)(\mathbf{v_p}) = \sum_i f_i(\mathbf{p})\, dx_i(\mathbf{v}) = \sum_i f_i(\mathbf{p}) v_i. \tag{10}$$

5. The Effect of d on the Products and Compositions of Functions

Definition 5.1. The *differential operator* d is a linear mapping that transforms a smooth real-valued function $f : \mathbf{E}^3 \to \mathbb{R}$ into a 1-form df.

For any point $\mathbf{p} \in \mathbf{E}^3$ and any tangent vector $\mathbf{v_p}$, the operator satisfies:

$$df(\mathbf{v_p}) = \mathbf{v_p}[f] \tag{11}$$

where $\mathbf{v_p}[f]$ is the directional derivative of f in the direction of $\mathbf{v_p}$.

In standard coordinates (x_1, x_2, x_3), the action of the operator on a function can be written explicitly as:

$$df = \sum_{i=1}^{3} \frac{\partial f}{\partial x_i} dx_i \tag{12}$$

Lemma 5.2. *Let fg be the product of differentiable functions f and g on $\mathbf{E}^3$. Then*

$$d(fg) = g\, df + f\, dg. \tag{13}$$

Proof. Using Corollary 5.5, we obtain

$$d(fg) = \sum_i \frac{\partial (fg)}{\partial x_i} dx_i = \sum_i \left(\frac{\partial f}{\partial x_i} g + f \frac{\partial g}{\partial x_i}\right) dx_i \tag{14}$$

$$= g\left(\sum_i \frac{\partial f}{\partial x_i} dx_i\right) + f\left(\sum_i \frac{\partial g}{\partial x_i} dx_i\right) = g\, df + f\, dg. \tag{15}$$

□

Lemma 5.3. *Let $f : \mathbf{E}^3 \to \mathbf{R}$ and $h : \mathbf{R} \to \mathbf{R}$ be differentiable functions, so the composite function $h(f) : \mathbf{E}^3 \to \mathbf{R}$ is also differentiable. Then*

$$d(h(f)) = h'(f)\, df. \tag{16}$$

Proof. (The prime here is just the ordinary derivative, so $h'(f)$ is again a composite function, from $\mathbf{E}^3$ to $\mathbf{R}$.) The usual chain rule for a composite function such as $h(f)$ reads

$$\frac{\partial(h(f))}{\partial x_i} = h'(f)\frac{\partial f}{\partial x_i}. \tag{17}$$

Hence

$$d(h(f)) = \sum_i \frac{\partial(h(f))}{\partial x_i} dx_i = \sum_i h'(f)\frac{\partial f}{\partial x_i} dx_i = h'(f)\, df. \tag{18}$$

□

The usefulness about lemmas 5.2 and 5.3 (along with their respective proofs), is that df reveals all partial derivatives of f, and *all of its directional derivatives.*

Example 5.4. For example, suppose

$$f = (x^2 - 1)y + (y^2 + 2)z. \tag{19}$$

Then by Lemmas 5.2 and 5.3,

$$df = (2x\,dx)y + (x^2 - 1)\,dy + (2y\,dy)z + (y^2 + 2)\,dz \tag{20}$$

$$= \underbrace{2xy}_{\partial f/\partial x}\ dx + \underbrace{(x^2 + 2yz - 1)}_{\partial f/\partial y}\ dy + \underbrace{(y^2 + 2)}_{\partial f/\partial z}\ dz \tag{21}$$

Now use the rules above to evaluate this expression on a tangent vector $\mathbf{v}$. The result is

$$\mathbf{v_p}[f] = df(\mathbf{v_p}) = 2p_1p_2v_1 + (p_1^2 + 2p_2p_3 - 1)v_2 + (p_2^2 + 2)v_3. \tag{22}$$

REFERENCES

[1] B. O'Neill, *Elementary Differential Geometry*, Revised 2nd ed. Boston, MA: Academic Press, 2006.